

The LCD2004A character dot matrix LCD display module has the advantages of small size, light weight, and high performance

Widely used due to its low consumption and rich display content, it utilizes liquid crystals that have been processed

Can change the direction of light propagation and stimulate liquid crystal molecules with electric current to produce dots, lines, and surfaces in combination

Combine the backlight to form the image. The LCD controller initially used in the LCD2004 module is

HD44780, And various LCD manufacturers provide almost all 1602 models with the same specifications

Blocks or compatible modules mostly use ICs that are compatible with them, so they are based on their characteristics

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It's actually the same; Of course, many manufacturers offer display modules with different character colors, backlight colors, and so on.

Deshengtai Technology

The interior of LCD2004 mainly consists of an LCD display screen (LCD PANEL) and a controller (control)-

ler)、The driver and bias voltage generation circuit are composed.

Display screen	Blue screen, green screen
temperature	-10~+60℃
controller	SPLC780 Technology
Storage temperature	-20~+70℃
size	0.55mm × 0.55mm backlight type EL/LED
Display content	20 characters x 4 lines, working voltage+5V
Module size	98mmx60mm
connection method	Conductive adhesive strip

8. Device address: It can be modified by short circuiting A0/A1/A2, and the default address can be detected using the program

Address modification instructions: (Taking the default address 0x27 as an example)

A0, A1, A2 are all suspended, and the device address is 0x27

Short circuit A0, device address 0x26

Short circuit A1, device address 0x25

Short circuit A2, device address 0x23

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A0, A1, A2 are all short circuited, and the device address is 0x20

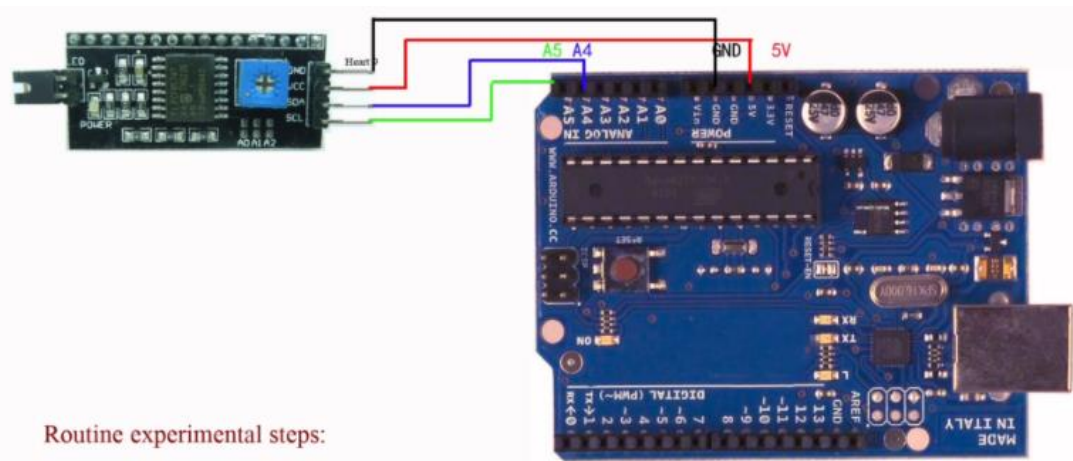
Wiring instructions:

GND: Negative pole of system power supply

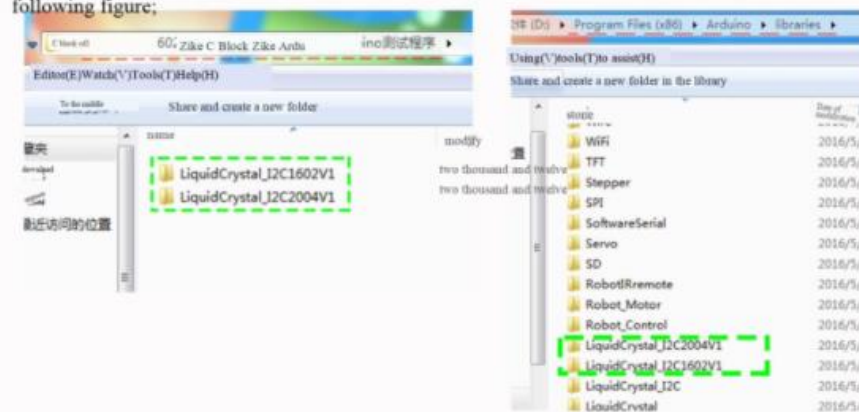
VCC: Positive pole of system power supply

SDA: IIC Data Cable

SCL: IIC clock line

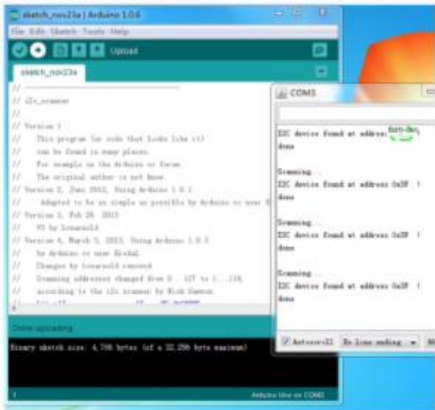


1. Install the library files and copy the folders from the supporting materials to the location shown in the following figure;



2. The default starting address of the detection module (run the address detection program in the download package,

Retrieve the serial monitor and read the default starting address;



The screenshot shows the Arduino IDE interface. The main window displays a C++ sketch named 'sketch_res12a' for an Arduino Uno. The sketch includes comments in Chinese and a line of code: `uint8_t _pin = 12;`. The serial monitor window is open, showing the output of the program. It displays three lines of text: 'I2C device found at address: 0x50 (0x)', 'I2C device found at address: 0x51 (0x)', and 'I2C device found at address: 0x52 (0x)'. The status bar at the bottom indicates 'Arduino Uno as COM5'.

3. Open the routine in the installed library according to the following path, change the address marked in yellow to the address you have detected, compile and download, and then restart the 1602 LCD to see the text display.

